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Episode 10 · September 22, 2026 · 2 papers

Leader-Follower Formation Control with Prescribed Convergence Rates under Bearing Persistence of Excitation

[SKIM](#)

Tarek Bouazza et al.

They replace the fixed gain in bearing-only leader-follower formation control with a gain matrix that adapts to the bearings each follower actually sees, so you can dial in the exponential convergence rate instead of inheriting whatever rate the formation's excitation happens to give you.

PROBLEM In bearing-based formation control with fixed gains, the guaranteed convergence rate is hostage to the persistence of excitation bound of the desired formation, so a formation geometry that is only weakly exciting converges slowly and you cannot fix it by tuning. The gap was that nobody had decoupled the achievable rate from that excitation bound while keeping the gain well posed and bounded.

RESULTS The abstract reports no numbers at all. There is no convergence rate figure, no noise level, no formation size, and no benchmark beyond a qualitative simulation comparison against fixed-gain designs, across three figures in eight pages.

METHOD Each follower runs a time-varying matrix gain driven by its own accumulated relative bearing measurements, in the spirit of the excitation-normalizing gains used in adaptive estimation. They prove the gain is well posed and uniformly bounded whenever the formation is bearing persistently exciting, and show the closed loop converges exponentially at a rate set by one tunable design parameter rather than by the excitation bound. The design is then carried over from single-integrator to double-integrator follower dynamics.

CAVEAT The whole evidence base is simulation against the authors' own fixed-gain baseline, with no measurement noise model stated and no hardware. The sharper concern is conditioning: a gain built by accumulating bearing information must be inverted or normalized somewhere, and uniform boundedness under ideal bearing persistence of excitation says nothing about what happens when excitation is marginal, when bearings are noisy, or when a follower briefly loses sight of its leader. No ablation on how the tunable rate parameter trades against control effort or actuator saturation, which is exactly where a prescribed-rate claim usually breaks.

WHY IT MATTERS This is peripheral to your work and you should treat it as such. It is sensor-only formation keeping with no communication in the loop at all: no intent exchange, no proposals or acceptances, no negotiation, and a fixed leader-follower hierarchy that assumes away the question of who yields. The one thread worth noting is that it lives on the opposite side of your central dichotomy, since it shows how far you can get on relative measurements alone, which makes it a useful reference point when you argue why explicit maneuver coordination messages buy something that bearing feedback cannot. If you want the platooning connection, the double-integrator extension is the closest thing here to a string of vehicles, but the paper says nothing about string stability, communication loss, or partial participation.

arXiv:2609.19106v1 [PDF](#) [incremental](#)

Multi-Agent Event-Triggered LQG Control under Shared Communication Constraints

SKIM

Zahra Hashemi, Dipankar Maity

When a group of independent LQG plants share one channel with a hard cap on how many can transmit per step, the controllers still decouple agent by agent, and the only hard part left is deciding who gets to speak, which they solve with a mixed integer program and then approximate with a cheap auction.

PROBLEM Event-triggered control is usually derived for a single plant, where the trigger only trades that plant's estimation error against its own communication cost. Once several plants share a network with a per-step capacity limit, the triggering decisions become globally coupled even though the dynamics are not, and the standard separation results do not obviously survive that coupling.

RESULTS The abstract gives no numbers at all. It only asserts, qualitatively, that the model predictive scheduler gets the best trade-off between control cost and communication cost, and that the auction scheduler lands close to it at substantially lower computational cost, with no agent counts, horizon lengths, solve times or percentage gaps stated.

METHOD They prove the optimal control law stays certainty-equivalent and splits into one independent finite-horizon Riccati recursion per agent, so the coupling lives entirely in the schedule. The schedule is then handled by a centralized receding-horizon scheduler, written as a mixed-integer linear program using a closed-form expression for the estimation-error covariance under a given transmission pattern. Two scalability devices sit on top: a window-based skip-pruning test that safely fixes runs of consecutive transmissions to zero before the solver runs, and an auction-inspired scheduler that ranks agents by a one-step transmission-benefit score.

CAVEAT Two things a reviewer would press on. First, everything rests on decoupled agent dynamics, so the shared network is the only thing tying the agents together; this is a bandwidth allocation result dressed as multi-agent control, and nothing here indicates what happens when the plants actually interact. Second, the scheduler is centralized and needs every agent's estimation-error covariance in one place at every step, which quietly assumes a perfect side channel for exactly the information the paper says is scarce. Add that the auction score is one step greedy with no stated optimality gap, and the claim that it comes close to the mixed-integer solution rests on unreported simulation numbers.

WHY IT MATTERS Peripheral. It shares your vocabulary, auctions and communication limits and degraded coordination, but not your problem: there is no negotiation here, no proposal and acceptance, no conflict to resolve, because the agents have nothing to agree about. The one transferable idea is the transmission-benefit score, which is a principled answer to the question of which vehicle should get the channel when maneuver coordination messages are congested, and that is worth borrowing as a congestion-control heuristic. But if you are looking for results on how coordination degrades under loss or partial participation, this paper assumes the capacity limit away into a scheduler rather than studying what breaks.

arXiv:2609.21974v1 PDF incremental